

Extended Abstract

Motivation Customized image generation remains a core challenge in text-to-image models, especially in scenarios where consistent subject identity must be preserved across diverse prompts, poses, and artistic styles. Existing solutions such as DreamBooth [22], LoRA [11], and IP-Adapter [31] require per-subject fine-tuning or rely on large vision encoders, making them inefficient or brittle when generalized to new conditions. Diffusion Self-Distillation (DSD) [3] provides a scalable alternative via self-supervised finetuning, but suffers from overfitting to its teacher’s limited generation manifold. As a result, it fails to generalize to compositional prompts, unfamiliar camera angles, or novel visual domains. We identify that subject-driven generation is fundamentally a preference alignment problem, rather than merely a reconstruction task, and propose a new training paradigm that directly optimizes generation toward preferred outputs.

Method We formulate subject-consistent image generation as a preference optimization problem and propose the first integration of Direct Preference Optimization (DPO) [20, 29] with Gaussian Mixture Flow Matching (GMFlow) [4], a structured diffusion model that models the reverse transition distribution as a multimodal Gaussian mixture. This design enables our model to better capture the rich diversity of valid image outputs while aligning them with user-intended subject identity and prompt semantics. To supervise DPO without requiring human feedback, we leverage vision-language models (VLMs) as automated reward oracles [18] to score subject consistency. Our system uses these scores to curate preference pairs, consisting of winning and losing samples, allowing us to apply the DPO objective in the space of GM transition densities.

Implementation We begin by initializing from FLUX-Fill and apply supervised finetuning (SFT) using the curated reference-target pairs from DSD [3], optimizing the GMFlow transition loss [4]. To collect data for DPO, we curate approximately 70k reference images from in-the-wild datasets using VLM-based filtering to ensure a clear main subject. For each reference-prompt pair, we generate four candidate images and score subject similarity using a VLM [18]. Pairs with clear preference contrast, such as excellent versus poor consistency, are retained, yielding approximately 12k high-quality training pairs. We adapt the Diffusion DPO objective [29] to our GMFlow framework, expressing the reward as a function of log-likelihood ratios over Gaussian mixtures. Only LoRA [11] and GM-specific layers are updated during DPO finetuning in order to avoid overfitting and maintain generalization.

Results We benchmark our method on the DreamBench++ dataset [18], which evaluates both concept preservation (CP) and prompt following (PF) via GPT-4o. Our method achieves a CP score of 0.808 with $\beta = 200$, outperforming the SFT baseline by 32% and the DSD model [3] by 28%. The combined metric CP-PF improves by 20% over the baseline and 36% over DSD. Visual inspection confirms that our model consistently retains subject identity, including shape, color, and facial features, while handling prompts with extreme stylistic variation and spatial transformations. Qualitatively, it succeeds across both photorealistic and non-photorealistic domains, indicating robust generalization.

Discussion Despite strong gains in subject fidelity, prompt following slightly decreases at low β due to the absence of prompt-aware supervision. A promising extension is to include a dual reward signal from VLMs, evaluating both subject consistency and prompt alignment, to further improve semantic controllability [18]. Our automated pipeline is highly scalable, and future work will explore larger datasets and more diverse preference sets, such as using eight to sixteen candidates per prompt, to refine training further. This framework also opens avenues for interactive or multi-turn preference alignment, enabling user-in-the-loop customization.

Conclusion We introduce a new framework for preference-aligned, subject-consistent image generation by integrating Diffusion DPO [29] with Gaussian Mixture Flow Matching [4]. Our model learns to prioritize samples that balance identity fidelity and prompt compliance by leveraging VLMs [18] as scalable reward oracles. This approach eliminates the need for manual tuning or handcrafted rewards, achieves state-of-the-art performance in compositional settings, and paves the way for scalable, controllable, and high-quality personalization in open-world generative tasks.

Gaussian Mixture Diffusion DPO for Customized Image Generation

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Abstract

We present a new framework for subject-consistent image generation that combines Direct Preference Optimization (DPO) [20, 29] with Gaussian Mixture Flow Matching (GMFlow) [4], enabling scalable, high-fidelity personalization without per-subject finetuning. Unlike prior approaches that rely on pixel-level reconstruction [3] or large vision encoders, we recast identity preservation as a preference alignment problem and optimize generation directly using vision-language model (VLM) scores as reward signals. Our method significantly outperforms supervised finetuning (SFT) and Diffusion Self-Distillation (DSD) [3], achieving a Concept Preservation score of 0.808 and a 20% gain in the CP-PF metric over SFT on the DreamBench++ benchmark [18]. Qualitatively, our model maintains subject identity across diverse prompts, styles, and poses, while generalizing to both photorealistic and stylized domains. This work demonstrates that preference-guided diffusion with structured Gaussian mixtures can serve as a powerful foundation for open-world, controllable image generation.

1 Introduction

Text-to-image diffusion models have revolutionized visual content creation, yet they remain frustrating tools for artists who need consistent control over a specific character or asset across many prompts and scenes. Even state-of-the-art personalization pipelines – DreamBooth [22], LoRA [11], IP-Adapter [31] variants – either demand per-instance training which is slow and GPU-heavy, or exhibit generalization robustness. Current personalization pipelines over-fit to the limited and often highly correlated reference views seen during fine-tuning, so identity fidelity collapses once the reference image and prompt departs from those conditions. Small perturbations such as novel camera angles, extreme poses, or unusual lighting frequently trigger attribute drift, causing the character’s facial proportions, color palette, or accessories to mutate unpredictably. Consequently, designers must either accept repetitive, near-duplicate outputs or invest additional GPU cycles to re-generate (even re-tune) for their desired subjects – an untenable workflow for large-scale, creative asset pipelines.

A recent work, Diffusion Self Distillation [3] (DSD) tackles personalization without external supervision by first sampling large batches of images from a frozen teacher diffusion model, then fine-tuning a student to reproduce those same samples, achieving zero-shot customized image generation without inference-stage tuning. However, because the entire training corpus is self-generated and therefore lies on the teacher’s narrow manifold, the student effectively learns to imitate the teacher’s existing

distribution rather than extrapolate beyond it. In practice, this closed-loop distillation amplifies any biases or mode collapse already present in the teacher and provides the student with no incentive to handle unseen poses, lighting conditions, or compositional prompts. Consequently, DSD [3] achieves impressive fidelity on images that resemble its own synthetic pre-training data, yet exhibits pronounced generalization failures, such as identity drift, background entanglement, and prompt ignoring, once the scene deviates from those self-sampled exemplars, making it ill-suited for open-world character customization scenarios.

A key observation is that subject-driven generation is fundamentally a preference-alignment problem: among the many valid samples the diffusion model can produce, humans prefer those that best match the reference subject while following the new prompt. Recent work on Direct Preference Optimization [20, 29] (DPO) shows that such preferences can be optimized directly by treating the model as a policy and pushing probability mass toward “winner” samples. Therefore, instead of naively optimizing pixel-space losses, which often yields poor generalization, we advocate training the diffusion model with Diffusion DPO [29] using the Gaussian Mixture Flow Matching (GMFlow) model [4] formulation, an extension of diffusion models that predicts Gaussian mixture denoising distributions. To our knowledge, this is the first work that integrates DPO [29] with GMFlow [4], leveraging the structured output space of Gaussian mixtures to better capture nuanced, user-specific preferences. This design enables fine-grained, preference-aware image generation without relying on handcrafted reward models or manual supervision., steering generation toward subject-faithful, prompt-compliant outputs and away from over-fitting to pixel-level details. Moreover, “subject consistency” is an inherently subjective judgment which is nuanced, context-dependent, and hard to capture with any hand-crafted metric. While designing a tractable analytic reward is infeasible, a preference-based objective such as DPO [29] naturally sidesteps this bottleneck by learning directly from relative preferences.

However, applying Diffusion DPO [29] to high-dimensional image manifolds poses another obstacle: collecting dense human comparisons is prohibitively expensive. We make our second key observation: state-of-the-art MLLMs/VLMs’ ability to identify whether two images share the same main subject is significantly better than any subject-driven image generation model, including DSD [3]. We therefore cast a VLM as a reward oracle: for sampled candidate images, the VLM supplies a consistency preference by comparing them with the reference, which we plug directly into the DPO [20, 29] objective. This strategy removes the need for costly human comparisons and captures the subtle semantic cues that hand-crafted metrics miss, giving Diffusion DPO [29] a powerful, scalable signal to align generation toward subject-consistent outputs.

We summarize our contributions as follows:

- We are the first to formulate subject-driven image generation as a preference-aligned learning problem, showing great generalization robustness across various inputs.
- We demonstrate that VLM-based rewards provide accurate, free, and scalable supervision for identity consistency, unlocking high-quality preference data without human effort.
- We are the first to formulate DPO [29] under GMFlow [4], leveraging the multimodal expressivity of GMFlow [4].

2 Related Work

Diffusion models. Diffusion probabilistic models [25, 10], score-based models [26, 27], and flow matching models [13, 14] formed a family of generative models that shared an underlying theoretical framework and had made significant advances in image generation [30, 21, 23, 19, 5, 7]. They defined a forward diffusion process that corrupted image data with Gaussian noise. The reverse denoising process was then modeled as a series of reverse transition distributions, each progressively removing a small amount of noise. Specifically, standard diffusion models approximated the denoising distribution as a Gaussian and trained neural networks to predict its mean. Subsequent works [16, 2, 1] additionally predicted the variance, and GMS [9] further extended this approach to third-order moments. More recently, Gaussian mixture flow matching (GMFlow) [4] introduced a more general approach that predicted a Gaussian mixture (GM) distribution of flow velocity, such that standard diffusion models became a special case with a single mixture component. By using more components, GMFlow was able to capture intricate multi-modal transitions and improve sample quality. However,

GMFlow only introduced a supervised training objective. The question of how to align GM predictions with human preferences remained unexplored.

Direct preference optimization. State-of-the-art large language models (LLMs)[17, 28] were typically finetuned after pretraining to enhance alignment with human preferences. Reinforcement learning from human feedback (RLHF) was a prominent alignment technique that involved first training a reward model on human preference datasets and subsequently using reinforcement learning (RL) to optimize the policy model. Rafailov et al.[20] introduced Direct Preference Optimization (DPO) as a simpler alternative, eliminating the need for a separate reward model by directly training the LLM on paired human preference data as an implicit reward model. The success of LLM DPO inspired Diffusion-DPO [29], which reformulated the DPO objective into a tractable bound expressed in terms of reverse transition distributions. Diffusion models finetuned using DPO demonstrated substantial improvements over their base counterparts in both visual quality and alignment with textual prompts. However, the original diffusion DPO objective was based on simplified transition distributions expressed in terms of residual noise (ϵ predictions), assuming simple Gaussian transitions with constant variances. In contrast, we explored DPO on the more expressive GM transition distributions.

Customized image generation. Subject-driven customized image generation aimed to synthesize novel images of a specific user-provided subject in diverse contexts, guided by text prompts. Early approaches focused on adapting pre-trained models through methods such as full fine-tuning (e.g., DreamBooth [22]) or optimizing textual embeddings (e.g., Textual Inversion [8]), often achieving high fidelity but requiring per-subject optimization and facing challenges in balancing subject preservation with prompt adherence. Subsequent work explored more efficient techniques like parameter-efficient fine-tuning (e.g., LoRA [11]) and encoder-based methods (e.g., IP-Adapter), which avoided test-time tuning but often relied heavily on encoder quality. Recently, Diffusion Self-Distillation (DSD) [3] introduced a novel zero-shot paradigm. DSD leveraged a pre-trained model’s inherent ability to generate consistent subject depictions in grids, used Vision-Language Models (VLMs) to automatically curate these into a large paired dataset, and then fine-tuned the model to become image+text conditional. Despite these advances, we observed that DSD still produced failure cases in which the generated images exhibited visual artifacts or failed to fully preserve the subject’s identity.

3 Preliminary

3.1 Gaussian Mixture Flow Matching Models

Let $\mathbf{x} \in \mathbb{R}^D$ be a data point sampled from a distribution with its PDF denoted by $p(\mathbf{x})$. A typical flow matching model defines a time-dependent interpolation between the data point and a random Gaussian noise $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, yielding the noisy data $\mathbf{x}_t = \alpha_t \mathbf{x} + \sigma_t \epsilon$, where $t \in [0, 1]$ denotes the diffusion time, and $\alpha_t = 1 - t$, $\sigma_t = t$ are the predefined time-dependent noise schedule. Apparently, the marginal PDF of the noisy data $p(\mathbf{x}_t)$ can be written as the data distribution $p(\mathbf{x})$ convolved by a Gaussian kernel $p(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \alpha_t \mathbf{x}_0, \sigma_t^2 \mathbf{I})$, i.e., $p(\mathbf{x}_t) = \int_{\mathbb{R}^D} p(\mathbf{x}_t|\mathbf{x}_0)p(\mathbf{x}_0) d\mathbf{x}_0$. Alternatively, $p(\mathbf{x}_t)$ can be expressed as a previous PDF $p(\mathbf{x}_{t-\Delta t})$ convolved by a transition Gaussian kernel $p(\mathbf{x}_t|\mathbf{x}_{t-\Delta t})$, given by:

$$p(\mathbf{x}_t|\mathbf{x}_{t-\Delta t}) = \mathcal{N}\left(\mathbf{x}_t; \frac{\alpha_t}{\alpha_{t-\Delta t}} \mathbf{x}_{t-\Delta t}, \overbrace{\left(\sigma_t^2 - \frac{\alpha_t^2}{\alpha_{t-\Delta t}^2} \sigma_{t-\Delta t}^2\right)}^{(\sigma_{t-\Delta t}^t)^2} \mathbf{I}\right), \quad (1)$$

where the transition variance is denoted by $(\sigma_{t-\Delta t}^t)^2$ for brevity. A series of convolutions from 0 to 1 constructs a diffusion process over time, establishing a mapping between $\mathbf{x}_0$ and $\mathbf{x}_t$ that can be described by a stochastic differential equation (SDE) or an ordinary differential equation (ODE) [27]. Flow matching models [13, 14] train a neural network to predict the ODE velocity field $\frac{d\mathbf{x}_t}{dt}$, so that the data can be generated by first sampling the noise ϵ and then solving the ODE to obtain $\mathbf{x}_0$. GMFlow [4] generalizes diffusion and flow matching models by modeling the reverse transition distribution $p(\mathbf{x}_{t-\Delta t}|\mathbf{x}_t)$ as a Gaussian mixture (GM):

$$q_\theta(\mathbf{x}_{t-\Delta t}|\mathbf{x}_t) = \sum_{k=1}^K A_k \mathcal{N}(\mathbf{x}_{t-\Delta t}; \boldsymbol{\mu}_k, s^2 \mathbf{I}), \quad (2)$$

where $A_k, \boldsymbol{\mu}_k, s$ are the GM parameters predicted by a neural network with weights θ , and K is a hyperparameter. This distribution is more informative than the ODE velocity alone, thus enabling high-

quality generation with GM-ODE and GM-SDE solvers [4]. The model can be further conditioned on text/image inputs $\mathbf{c}$, with the conditional reverse transition distribution denoted by $q_{\theta}(\mathbf{x}_{t-\Delta t}|\mathbf{x}_t, \mathbf{c})$. For practical image generation, GMFlow further factorizes the high-dimensional image distribution into the product of low-dimensional pixel-wise GM distributions (in the context of latent diffusion models [21], the factorization is conducted in the latent space).

3.2 Diffusion DPO

Let $q_{\theta}^{\text{SDE}}(\mathbf{x}_0|\mathbf{c})$ be the conditional PDF of data generated by a diffusion model using an SDE solver. Given a conditional reward function $r(\mathbf{x}_0, \mathbf{c})$, RLHF aims to finetune the model with the following objective:

$$\min_{\theta} -\mathbb{E}_{\mathbf{c}, \mathbf{x}_0 \sim q_{\theta}^{\text{SDE}}(\mathbf{x}_0)}[r(\mathbf{x}_0, \mathbf{c})] + \beta D_{\text{KL}}(q_{\theta}^{\text{SDE}}(\mathbf{x}_0|\mathbf{c}) \| q_{\text{ref}}^{\text{SDE}}(\mathbf{x}_0|\mathbf{c})), \quad (3)$$

where $q_{\text{ref}}^{\text{SDE}}(\mathbf{x}_0|\mathbf{c})$ represents the PDF of a reference model (e.g., the pretrained model), and β is a hyperparameter controlling the strength of Kullback–Leibler (KL) divergence [12] regularization. The optimal solution θ^* to the RLHF objective leads to the following distribution:

$$q_{\theta^*}^{\text{SDE}}(\mathbf{x}_0|\mathbf{c}) = \frac{q_{\text{ref}}^{\text{SDE}}(\mathbf{x}_0|\mathbf{c})}{Z(\mathbf{c})} \exp\left(\frac{r(\mathbf{x}_0, \mathbf{c})}{\beta}\right), \quad (4)$$

where $Z(\mathbf{c})$ is a condition-dependent normalization factor. Instead of using a reward function to finetune the model, DPO [20] formulates an implicit reward function in terms of the model by rewriting Eq. (4) into $r_{\theta}(\mathbf{x}_0, \mathbf{c}) = \beta \log \frac{q_{\theta}^{\text{SDE}}(\mathbf{x}_0|\mathbf{c})Z(\mathbf{c})}{q_{\text{ref}}^{\text{SDE}}(\mathbf{x}_0|\mathbf{c})}$. Given a dataset of winning and losing pairs $\mathbf{x}_0^w \succ \mathbf{x}_0^l|\mathbf{c}$, DPO trains the implicit reward using the Bradley-Terry preference model, with the following DPO loss function:

$$\mathcal{L}(\theta) = -\mathbb{E}_{\mathbf{c}, \mathbf{x}_0^w, \mathbf{x}_0^l}[\log \sigma(r_{\theta}(\mathbf{x}_0^w, \mathbf{c}) - r_{\theta}(\mathbf{x}_0^l, \mathbf{c}))], \quad (5)$$

where σ is the sigmoid function. In diffusion models, the PDF $q_{\theta}^{\text{SDE}}(\mathbf{x}_0|\mathbf{c})$ is intractable since it requires marginalizing out all the intermediate reverse transition distributions. Diffusion DPO [29] addresses this by replacing $q_{\theta}^{\text{SDE}}(\mathbf{x}_0|\mathbf{c})$ with the reverse transition PDF $q_{\theta}(\mathbf{x}_{t-\Delta t}|\mathbf{x}_t, \mathbf{c})$, which is a function of the predicted residual noise ϵ in the case of standard diffusion models.

3.3 Diffusion Self-Distillation

Diffusion Self-Distillation [3] (DSD) transforms a pretrained text-to-image diffusion model into a text+image-conditioned generator that performs zero-shot, identity-preserving customization with no test-time tuning. It leverages the teacher model’s “in-context grid” generation capability: grid prompts produced by an LLM coax the teacher to emit multi-panel images in which every panel depicts the same subject under varied poses, lighting, and scenes. A vision–language model (VLM) then filters the crops, automatically curating roughly 400k high-quality (reference, target) pairs without any human effort. The student, initialized from the teacher, is fine-tuned on these pairs via a two-frame “parallel-processing” architecture that treats the reference as frame 0 and the edited output as frame 1, allowing rich cross-attention without strict spatial alignment. Trained in this supervised fashion, DSD rivals DreamBooth [22] and LoRA [11] in concept fidelity, outperforms zero-shot baselines such as IP-Adapter [31] on prompt compliance and diversity, and retains instant inference speed since no further finetuning is needed. Its main limitation is that the curated pairs lie on the teacher’s own manifold, so the student inherits the teacher’s biases and can falter on out-of-distribution poses or compositions, motivating preference-aligned learning in this work.

4 Method

4.1 Supervised Finetuning

Building on the training scheme and data from DSD [3], we finetune FLUX-Fill into an in-context, inpainting-based custom image generator based on GMFlow [4]. We follow the original GMFlow and employ the GM transition loss for supervised finetuning (SFT):

$$\mathcal{L}(\theta) = \mathbb{E}_{\mathbf{x}, t, \mathbf{x}_{t-\Delta t}, \mathbf{x}_t}[-\log q_{\theta}(\mathbf{x}_{t-\Delta t}|\mathbf{x}_t, \mathbf{c})], \quad (6)$$

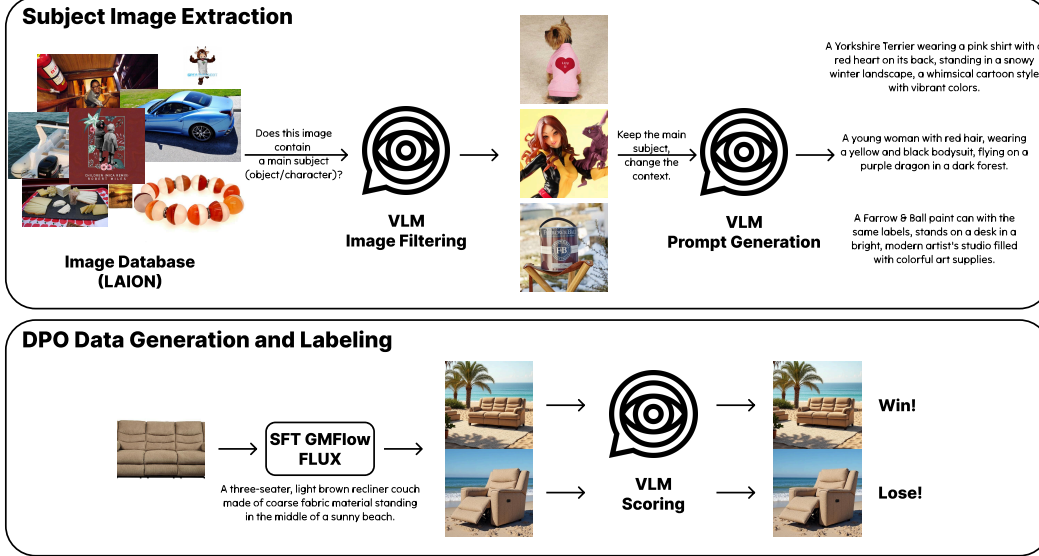


Figure 1: *Top*: We mine the LAION [24] dataset with a vision–language model (VLM) to (1) filter for images containing a single, clearly recognizable subject and (2) rewrite each image caption to place that subject into a new context, yielding diverse reference–prompt pairs with no human effort. *Bottom*: For each pair, our SFT GMFlow [4] FLUX model samples multiple candidates (only two shown for simplicity). The same VLM scores subject consistency, and we retain the best candidate image and a contrasting low scorer to form a DPO [29] preference pair. This fully automated loop supplies high-quality supervision for Direct Preference Optimization at scale.

where c denotes the input image and text, $x_{t-\Delta t}$, x_t are sampled by injecting noise into the ground truth output image, and θ is the LoRA [11] and GM output layer weights. The rest of the network weights are frozen to alleviate overfitting.

Unlike the original DSD pipeline which had to repurpose the text-to-image FLUX model into an image-conditioned architecture on the fly, we begin directly with FLUX-Fill, the officially released in-painting variant of FLUX. FLUX-Fill already encodes a strong spatial prior for masked-image completion and supports per-token cross-attention between the reference patch and the generation canvas. This removes the two most expensive steps in DSD: (i) synthesizing a “two-frame video” surrogate input during training and (ii) forcing the student to relearn low-level in-painting behavior from scratch. In practice we observe markedly faster convergence, because most of the backbone weights can remain close to their well-calibrated initialization. Because we retain DSD’s curated (reference, target) pairs, this “supervised full finetuning” step remains plug-and-play: simply swap in FLUX-Fill as the student backbone, replace the flow-matching loss with the GM transition loss in Eq. (6), and the training could converge within 2000 iterations. The result is an in-context custom generator that inherits FLUX-Fill’s strong reconstruction prior and GMFlow’s multi-modal expressivity, without the compute overhead or architectural contortions required by the original DSD [3] design.

4.2 DPO Data Collection

Fig. 1 shows our DPO [20, 29] data collection process. Specifically, we first collect images where clear main subjects can be defined, pair each with a context-changing prompt (Fig. 1, top), then generate multiple DPO [20, 29] candidates which are scored by a VLM on their consistency with the reference, from which we filter out the (win, lose) pairs (Fig. 1, bottom) ready for DPO [20, 29] finetuning.

Source Subject Images Extraction. Large web-scale image datasets like LAION are diverse, but most images in them are unusable for subject-driven customization: many contain multiple people, tiny foreground objects lost in scenery, heavy text overlays, collages, memes, screenshots, or purely abstract patterns. Training on such noise would teach the model to “preserve” clutter rather than a single, recognizable protagonist. We therefore insert a VLM-based subject filter before any data reaches the generator. For every candidate image, we prompt the vision–language model with a

yes/no query – “Does this picture feature one clear main subject (object or character) that can be referred to unambiguously?”, and discard anything that fails the test. The filter eliminates scenes with ambiguous focus, crowded group photos, and emblem-style graphics, yielding a clean pool of images where a single dog, gadget, toy, person, or product dominates the frame. This curated subset forms the backbone of our pipeline.

DPO Candidate Generation. With each reference image having an unambiguous focal entity, we can safely use the VLM again to write rich, context-shifting prompts that preserve the subject description verbatim while replacing the surrounding scene, style, lighting, or activity (e.g., “A Yorkshire Terrier wearing a pink shirt...*in a snowy winter landscape*”). The resulting (reference-image, rewritten-prompt) pairs give us a diverse catalogue of subjects—pets, people, gadgets, food, toys—each decoupled from its original background and ready for downstream DPO [20, 29] training, all without manual annotation. For each reference image and its subject-preserving prompt, we query the SFT model to generate several independent samples (typically four) by changing the random seed. Generating a small batch instead of just a pair of candidate image is crucial: many prompts yield either all poor results (all losing) or a set of perfectly acceptable images (all winning), and both situations fail to create the contrast that DPO [20, 29] needs. The more candidates we generate per reference-prompt pair, the higher the likelihood that at least one image will clearly be the “winning” sample and one will clearly be the “losing” sample, ensuring a useful preference signal.

DPO Pair Curation and Labeling. Once a batch of candidate images is generated, we query the VLM to rate the subject-consistency of each candidate image with the reference image on an ordinal scale from 0 (poor) to 4 (perfect). For every reference-prompt pair we then keep the highest-scoring candidate (score = 4) as the winner and the lowest-scoring candidate (score ≤ 2) as the loser; if a batch lacks either a 4-point image or any image scoring ≤ 2 , we discard the entire set. This simple filtering step ensures that every training example supplies a meaningful preference contrast while maintaining fully automated, scalable labeling. We design our scoring prompt following evaluation benchmark DreamBench++ [18]. The rubric (full text in the supplementary) instructs the model to compare the two main subjects, whether a character, animal, or object, along four explicit visual axes: shape (global outline and proportions), color (hue, saturation, brightness distribution), texture (local detail fidelity), and, when relevant, facial features (eyes, muzzle, etc.). The VLM must return a single integer score from 0 to 4: Very Poor (0, no resemblance), Poor (1), Fair (2), Good (3), or Excellent (4, near-identical), and is instructed to assign a hard 0 whenever no clear central subject is detectable or when border artifacts such as stray vertical crops or edge bars are present.

4.3 Gaussian Mixture Diffusion DPO

With the curated winning–losing pairs, we apply DPO finetuning to the SFT model. To leverage GMFlow’s expressive reverse transition distribution, we adapt the diffusion DPO [29] approach to GMFlow by expressing the DPO objective (Eq. (5)) in terms of GM transitions, thus establishing a novel Gaussian mixture DPO loss function:

$$\mathcal{L}(\theta) = -\mathbb{E}_{\mathbf{c}, t, \mathbf{x}_{t-\Delta t}^w, \mathbf{x}_{t-\Delta t}^l, \mathbf{x}_t^w, \mathbf{x}_t^l} \left[\log \sigma \left(\beta \log \frac{q_{\theta}(\mathbf{x}_{t-\Delta t}^w | \mathbf{x}_t^w, \mathbf{c})}{q_{\text{ref}}(\mathbf{x}_{t-\Delta t}^w | \mathbf{x}_t^w, \mathbf{c})} - \beta \log \frac{q_{\theta}(\mathbf{x}_{t-\Delta t}^l | \mathbf{x}_t^l, \mathbf{c})}{q_{\text{ref}}(\mathbf{x}_{t-\Delta t}^l | \mathbf{x}_t^l, \mathbf{c})} \right) \right], \quad (7)$$

where $\mathbf{c}$ is the conditioning signal (input text and image), $\mathbf{x}_{t-\Delta t}^w, \mathbf{x}_{t-\Delta t}^l, \mathbf{x}_t^w, \mathbf{x}_t^l$ are noisy samples created from a winning–losing pair using the forward diffusion process, and $q_{\theta}, q_{\text{ref}}$ represent the probability density of the DPO-finetuned transition and the reference SFT transition, respectively. Under the GMFlow framework, all the transitions are expressed as GM PDFs (Eq. (2)). In contrast, the original diffusion DPO expresses the transitions in terms of residual noises (ϵ in Eq. (14) in [29]). Compared to the original diffusion DPO, our expressive GM transition can better approximate the marginal distribution $q_{\theta}^{\text{SDE}}(x_0 | \mathbf{c})$ of the data generated using the SDE solver, thus our GM DPO objective is closer to the exact objective in Eq. (5).

Same as the SFT stage, we finetune only the LoRA [11] and GM output layer weights for DPO, while freezing the rest of the network weights to alleviate overfitting. A more detailed description of the training method is presented in Algorithm 1.

Algorithm 1: GM DPO training scheme.

Input: DPO dataset $\{x_0^w, x_0^l, c\}$, SFT model q_{ref} , regularization param β , shift a , transition ratio λ

Output: Network params θ

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1  $\theta \leftarrow$  SFT model weights
2 for sample  $x_0^w, x_0^l, c$  do
3   Sample  $t \sim \text{LogitNormal}(0, 1)$  // Logit-normal sampling proposed in SD3 [7]
4    $t \leftarrow \frac{at}{1+(a-1)t}$  // Time shifting proposed in SD3 [7]
5    $\Delta t \leftarrow \lambda t$  // GMFlow transition schedule [4]
6   Sample  $\epsilon_1 \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ ,  $\epsilon_2 \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
7    $x_{t-\Delta t}^w \leftarrow \alpha_{t-\Delta t} x_0^w + \sigma_{t-\Delta t} \epsilon_1$ ,  $x_{t-\Delta t}^l \leftarrow \alpha_{t-\Delta t} x_0^l + \sigma_{t-\Delta t} \epsilon_1$ 
8    $x_t^w \leftarrow \frac{\alpha_t}{\alpha_{t-\Delta t}} x_{t-\Delta t}^w + \sigma_{|t-\Delta t}^t \epsilon_2$ ,  $x_t^l \leftarrow \frac{\alpha_t}{\alpha_{t-\Delta t}} x_{t-\Delta t}^l + \sigma_{|t-\Delta t}^t \epsilon_2$  // Eq. (1)
9    $\mathcal{L} \leftarrow -\log \sigma \left( \beta \log \frac{q_\theta(x_{t-\Delta t}^w | x_t^w, c)}{q_{\text{ref}}(x_{t-\Delta t}^w | x_t^w, c)} - \beta \log \frac{q_\theta(x_{t-\Delta t}^l | x_t^l, c)}{q_{\text{ref}}(x_{t-\Delta t}^l | x_t^l, c)} \right)$ 
10  Backpropagate and update  $\theta$ 

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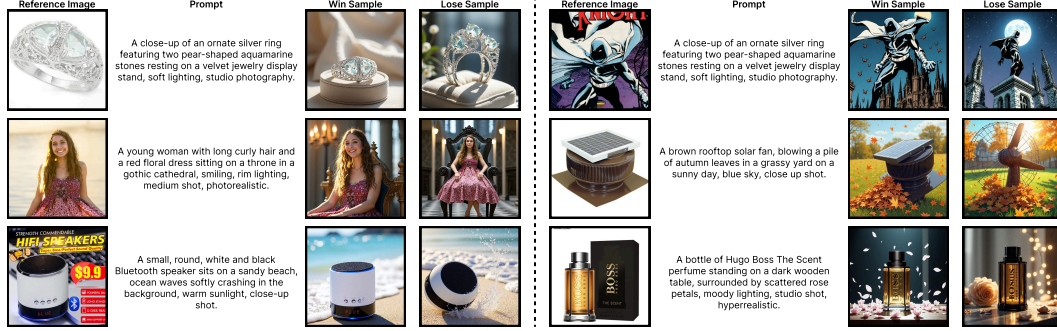


Figure 2: Exemplar image pairs from our Diffusion DPO [29] dataset.

5 Experiment Setup

Implementation details. We use FLUX-Fill as our base model, and finetune on the DSD [3] data to obtain our improved GMFlow-based SFT model, then perform Diffusion DPO [20, 29] on this SFT model using different β . For GMFlow [4] hyperparameters, we follow the original paper and set $K = 8$, $\lambda = 0.5$. We use Gemini-2.0-Flash across our setup as the LLM/MLLM/VLM, because of its high performance and relatively low cost. We finetune all models on NVIDIA H100 80 GB GPU nodes for 2000 iterations with an effective batch size of 160 for SFT or 256 for DPO, using 8-bit AdamW optimizer [15, 6] with a learning rate of 10^{-4} . We use LoRAs [11] with rank 128 across our finetuning experiments. For inference, we generate samples using the GM-ODE 2 solver [4] with 20 time steps.

DPO Dataset. We sample around $\sim 70k$ subject images in total, where each image is paired with a target context-changing prompt. For each (reference image, prompt) pair, we use our SFT model to generate 4 potential DPO candidates, and use the VLM to score their consistency with the reference image. After filtering, we obtain $\sim 12k$ distinct image pairs with clear winning and losing samples to use for Diffusion DPO [29]. Exemplar images of our DPO dataset is shown in Fig. 2.

Evaluation. Following DSD [3], we use the publicly available DreamBench++ [18] dataset and follow their protocols for evaluation. DreamBench++ [18] is a comprehensive and diverse dataset for evaluating personalized image generation, consisting of 150 high-quality images and 1,350 prompts, which are significantly more than previous benchmarks like DreamBench [22]. The dataset covers various categories such as animals, humans, objects, etc., including photorealistic and non-photorealistic images, with prompts designed to span different difficulty levels (simple/imaginative). We note that GPT-4o is used to automatically score generated images on two key aspects: Concept Preservation (how well the subject’s identity is maintained) and Prompt Following (how well the image adheres to the text prompt).

Table 1: **Quantitative comparison with Ablation** on the regularization strength β .

Method	Concept Preservation $\uparrow$	Prompt Following $\uparrow$	CP·PF $\uparrow$
Original DSD [3]	0.631	0.726	0.458
SFT Baseline	0.611	0.847	0.518
DPO $\beta = 200$	0.808	0.769	0.621
DPO $\beta = 1000$	0.735	0.813	0.598
DPO $\beta = 2000$	0.643	0.843	0.542

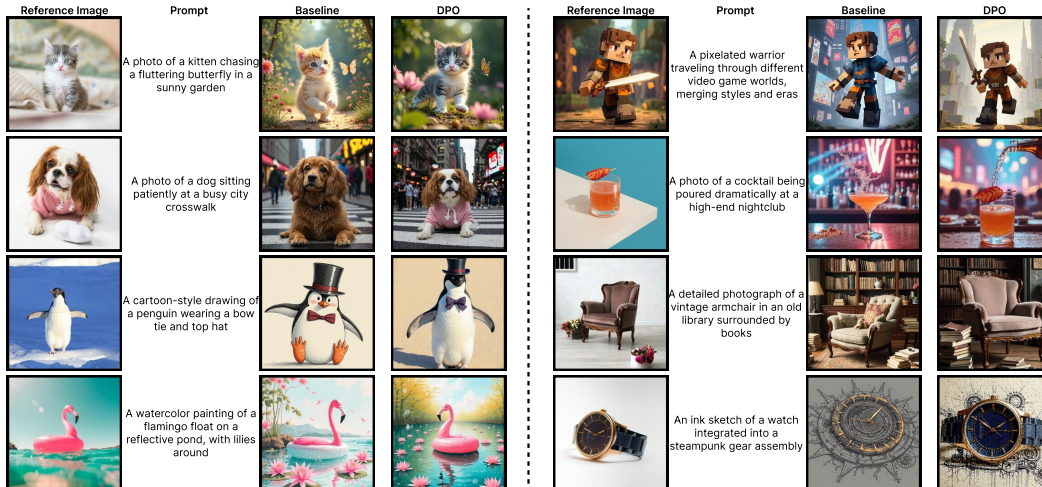


Figure 3: **Qualitative comparison** between our baseline SFT model and DPO model with $\beta = 1000$.

6 Results

6.1 Quantitative Results

Tab. 1 summarizes how DPO [20, 29] affects our SFT model as we sweep the regularization coefficient β . We note that a smaller β lets DPO move farther from the SFT baseline, whereas a larger β keeps the policy close to its original weights. With $\beta = 200$, the *Concept Preservation* (CP) score climbs to 0.808, representing a 32% gain over the SFT Baseline and a 28% gain over the original DSD [3], confirming that preference-based optimization effectively strengthens identity fidelity. As β increases, CP gradually decays to 0.735 at $\beta = 1000$ and 0.643 at $\beta = 2000$, mirroring the stronger pull toward the pre-DPO weights. In contrast, *Prompt Following* (PF) behaves in the opposite direction: because our DPO [20] pairs are labeled solely by subject consistency, aggressive updates slightly reduce PF (down to 0.769 at $\beta = 200$), but the score recovers and even surpasses the baseline (0.847) when β is large enough to leave the SFT behavior mostly intact. The overall metric given by CP·PF peaks at $\beta = 200$ with a value of 0.621, outperforming the baseline by 20% and DSD by 36%. Larger β values erode this advantage: the regained prompt adherence cannot offset the loss in concept fidelity. These trends highlight that a moderate regularization setting ($\beta \approx 1000$) offers the best balance, delivering strong subject consistency while maintaining competitive prompt alignment. While our current DPO [20, 29] setup focuses exclusively on subject-consistency, the results suggest that incorporating prompt adherence into the reward would further boost overall quality.

6.2 Qualitative Results

Fig. 3 provides a side-by-side visual comparison that reinforces these observations. Across every row—spanning realistic photographs, low-poly game characters, watercolor scenes, and minimalist sketches—the baseline model either drifts from the reference (altered body proportions, missing textures) or replicates it with minimal variation. In contrast, the DPO [20, 29] result consistently preserves the salient shape, color palette, and fine-grained detail of the original subject while cleanly injecting the new scene or stylistic cue prescribed by the prompt. The improvement is apparent for both animate (animals, stylized characters) and inanimate (furniture, product shots)

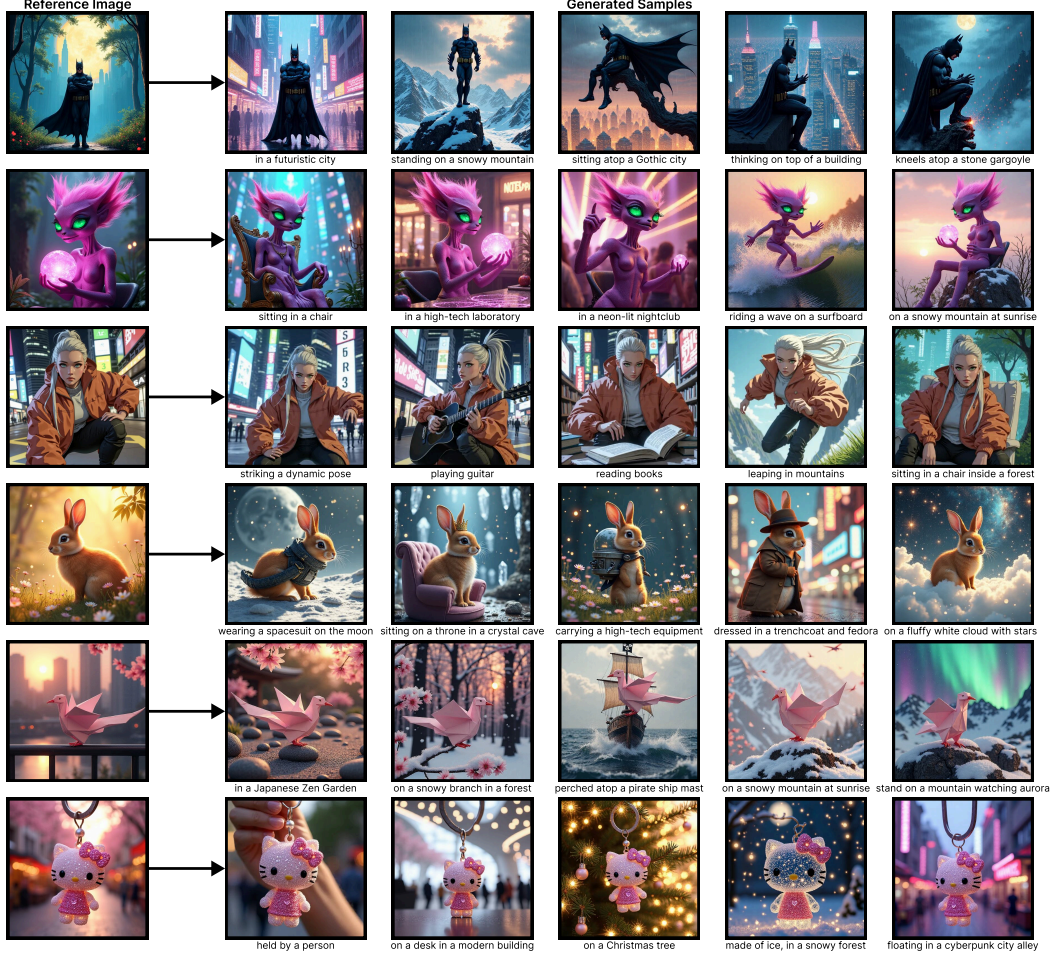


Figure 4: **Qualitative results** on various input images and context-changing target prompts.

categories, indicating that preference alignment scales gracefully across domains and rendering styles. Qualitative inspection shown in Fig. 4 further confirms that the DPO [20, 29]-trained model delivers tight identity consistency, even as prompts push the subject through extreme changes in scene, pose, lighting, or artistic style. Crucially, these performances are not confined to any single domain: the model preserves characters and objects with equal reliability in photorealistic renders, painterly illustrations, cell-shaded cartoons, and highly stylized “neon-cyberpunk” or “oil-on-canvas” settings. The fidelity persists across prompt categories, from action verbs and camera-angle cues to multi-sentence, narrative descriptions, demonstrating that DPO [29] generalizes robustly to long, compositional inputs.

Fig. 5 further reveals the impact of the regularization strength β on DPO finetuning. When $\beta = 5000$, the strong regularization diminishes the advantage of DPO finetuning, leading to subpar concept fidelity. When $\beta = 200$, the DPO finetuning is highly aggressive and produces results with highly consistent subjects, but we observe that the image quality is degraded and patch artifacts are noticeable. $\beta = 1000$ offers the most balanced results, aligning with the quantitative results in Tab. 1.

In summary, preference alignment not only sharpens subject retention but also scales gracefully to the full breadth of styles, domains, and prompt complexities encountered in real-world customization workflows.

7 Discussion

Limitation and Future Work. Our current reward does not take into account the prompt following criteria, which is a direct reason why our DPO [20, 29] model suffers from Prompt Following metric

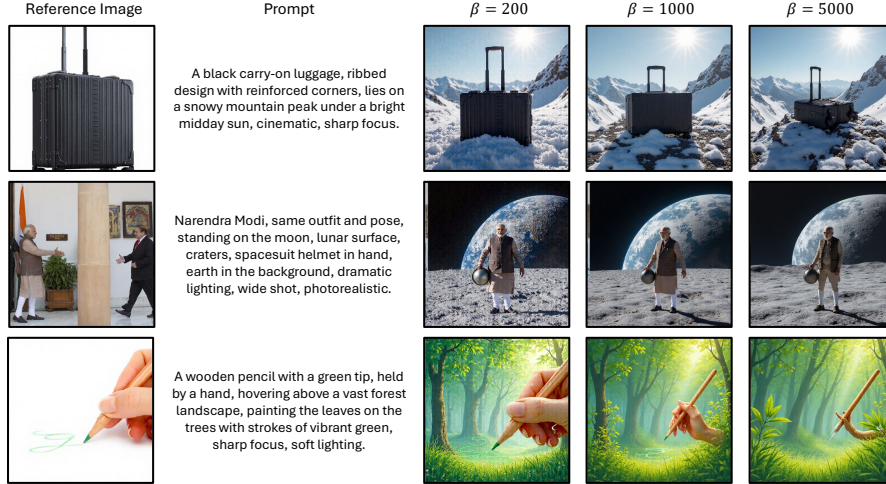


Figure 5: **Impact of regularization strength β on DPO models.**

drop. Therefore, a straightforward extension is to ask the VLM for two independent scores, one for “same-subject?” and one for “prompt-match?”, and combine them in a weighted sum or curriculum schedule. This dual-objective signal would encourage the model to keep the reference identity *and* satisfy scene, style, or composition cues, removing the mild Prompt Following degradation we observe at low β . Because modern VLMs already excel at both consistency and instruction following, adding a prompt component requires no human annotation and slots naturally into the existing DPO pipeline; preliminary experiments indicate that even a modest 20-30% weight on prompt following can recover most of the lost Prompt Following score without eroding the Concept Preservation gains. In addition, our data budget is still modest. We extract roughly 70K reference subjects and generate only four DPO [20, 29] candidates per prompt, yielding only roughly 12K pairs for DPO [20, 29]. Our data pipeline is fully automated, so given sufficient time, scaling to *millions* of subjects and sampling much more (e.g., $8-16$) candidates per prompt is straightforward and should produce richer winner-loser contrasts, further boosting both fidelity and robustness. We leave this large-scale exploration, along with integrating prompt-aware rewards and investigating multi-turn interactive feedback, as promising directions for future work.

Broader Impact. By removing per-subject fine-tuning, our approach lowers the barrier for artists and small studios to create consistent characters across large prompt libraries; it can also facilitate educational or assistive content generation for users with limited design skills. At the same time, easier identity preservation raises familiar misuse risks, such as producing convincing deepfakes, infringing on copyrighted characters, or amplifying bias present in our dataset and VLM reward signals. We are mindful of the potential misuse, particularly in deepfakes. We oppose exploiting our work for purposes that infringe upon ethical standards or privacy.

8 Conclusion

We introduced the first framework that combines Direct Preference Optimization [20, 29] with Gaussian Mixture Flow Matching [4] for subject-driven image generation. By casting the task as a preference alignment problem and leveraging vision-language models as semantic reward oracles, our method achieves strong identity preservation and generalization without manual labeling or per-subject tuning.

This integration enables scalable and high-quality customization across a wide range of prompts and visual domains. Our experiments show that the proposed method consistently outperforms prior approaches, including Diffusion Self-Distillation (DSD) [3], especially in compositional and out-of-distribution scenarios.

More broadly, this work defines a new direction for controllable image generation. It provides a structure-aware, preference-aligned, and fully automated pipeline that supports open-world subject

customization. These insights open opportunities for more robust, semantically grounded, and user-aligned generative models in the future.

9 Team Contributions

- **Shengqu Cai** led the implementation and analysis of the Diffusion Self-Distillation (DSD) framework. This included constructing the identity-consistent image grid generation pipeline, applying VLM-based curation for automatic preference labeling.
- **Hansheng Chen** was responsible for the design, implementation, and optimization of the Gaussian Mixture DPO model. He integrated GMFlow with the DPO objective and conducted model training and parameter tuning.
- **Lei Shu** managed experiment results, selected comparison images, and led the evaluation and analysis pipeline. Additionally, she contributed to the final report, paper writing, poster design, and presentation preparation.

Changes from Proposal. The original hypothesis stated that only the positive and negative pairs from the DSD [3] data generation and curation pipeline would be used. This was later revised to include the collection of model inference results using other in-the-wild images, which were then labeled as positive or negative by GPT-4o. This adjustment aligned with our findings that out-of-distribution samples impair the effectiveness of Diffusion DPO [29]. The supervised fine-tuning stage leveraged only the winning samples, which could already push the model close to its optimal distribution. As a result, the losing samples used in Diffusion DPO lay largely out-of-distribution, potentially limiting the method’s capacity to deliver further gains. To address this, we leveraged in-the-wild images to generate additional outputs and used GPT-4o or Gemini to automatically label each pair as positive or negative. The inclusion of GPT-labeled in-the-wild samples provided a more nuanced preference landscape, ultimately improving model adaptability.

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A Full DPO Candidate Scoring Prompt

DPO Candidate Scoring Prompt

Task Definition
 You will be provided with two images.
 As an experienced evaluator, your task is to evaluate the semantic consistency between the main subject of the two images, according to the scoring criteria.
 The main subject could be a character, an animal or an item/object.

Scoring Criteria
 It is often compared whether two subjects are consistent based on four basic visual features:
 1. Shape: Evaluate whether the main body outline, structure, and proportions of the first image match those of the second image. This includes the geometric shape of the main body, clarity of edges, relative sizes, and spatial relationships between various parts composing the main body.
 2. Color: Comparing the accuracy and consistency of the main colors of the main subject in the first image with those of the second image. This includes saturation, hue, brightness, and whether the distribution of colors is similar to that of the main subject in the second image.
 3. Texture: Focus on the local parts of the RGB image, whether the first image effectively captures fine details without appearing blurry, and whether it possesses the required realism, clarity, and aesthetic appeal. Please note that unless specifically mentioned in the text prompt, excessive abstraction and formalization of texture are not necessary.
 4. Facial Features: If the evaluation is of a person or animal, facial features will greatly affect the judgment of image consistency, and you also need to focus on judging whether the facial area looks very similar visually.

Scoring Range
 You need to give a specific integer score based on the comprehensive performance of the visual features above, ranging from 0 to 4:
 - Very Poor (0): No resemblance. The two images' main subjects have no relation to each other.
 - Poor (1): Minimal resemblance. The main subjects fall within the same broad category but differs significantly.
 - Fair (2): Moderate resemblance. The main subjects shows likeness with notable variances.
 - Good (3): Strong resemblance. The main subject of the first image closely matches the main subject of the second image with only minor discrepancies.
 - Excellent (4): Near-identical. The main subjects of the two images are virtually indistinguishable.

Input format
 Every time you will receive two images.

Please carefully review each image of the main subject, and please be extremely critical.

Output Format
 Your Score (a single digit)

You must adhere to the specified output format, which means that only the integer scores need to be output, excluding your analysis process.

B Story Boarding

Similar to the original DSD [3], our model unlocks simple comic/manga/illustration creation, with even less cost than DSD [3]. We show an example story board in Fig. 6.

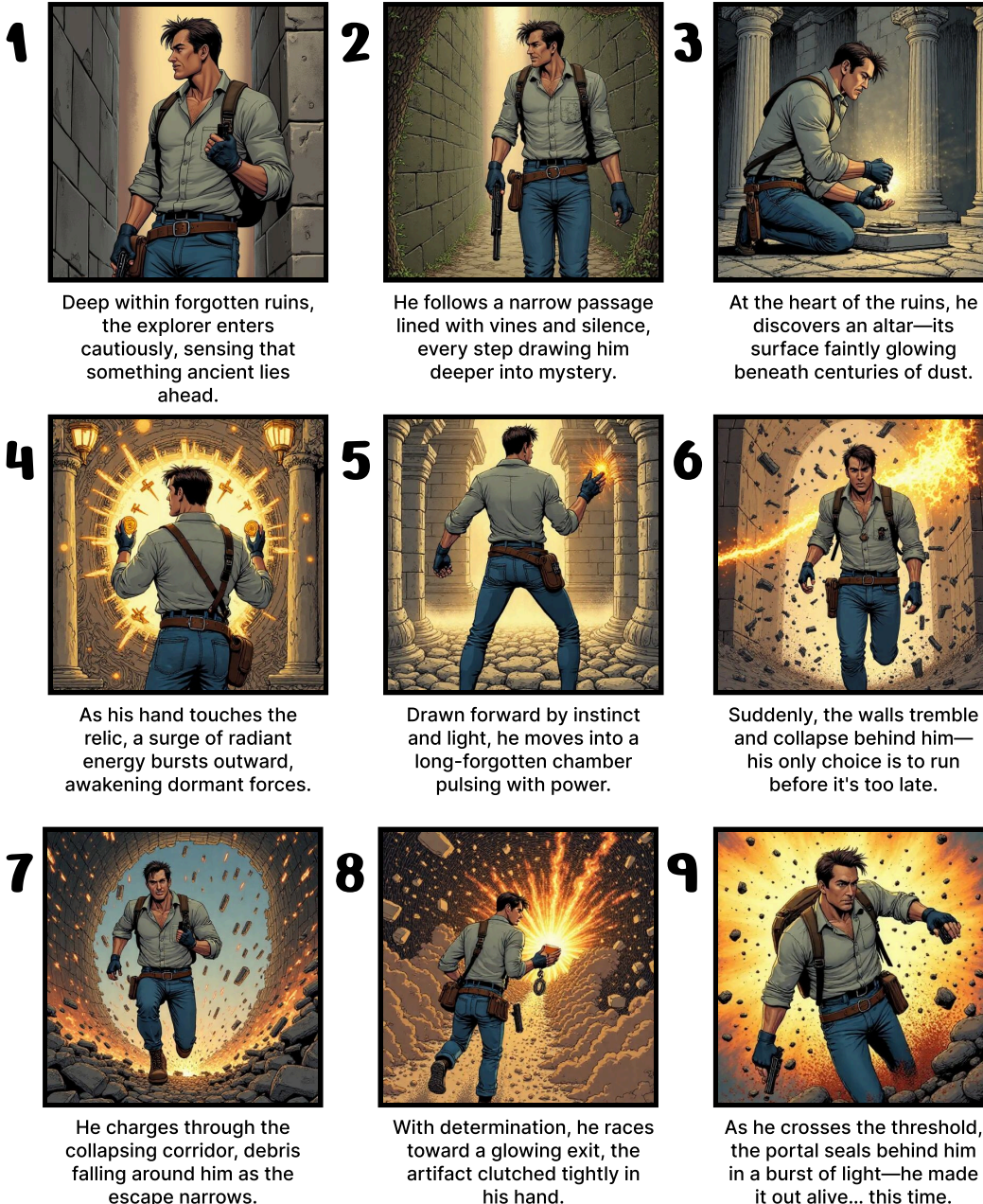


Figure 6: Simple storyboarding using our model.